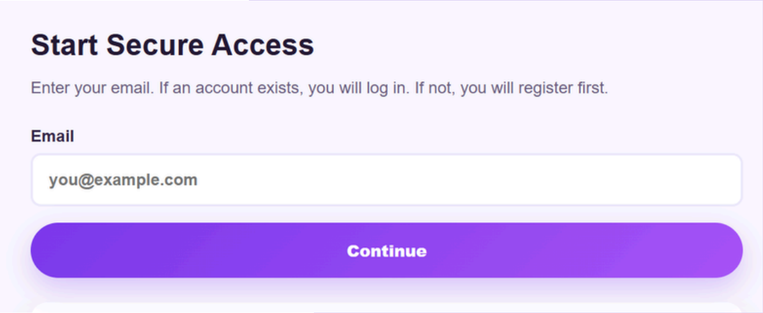
CJ Secure

Confidential Data Access Portal

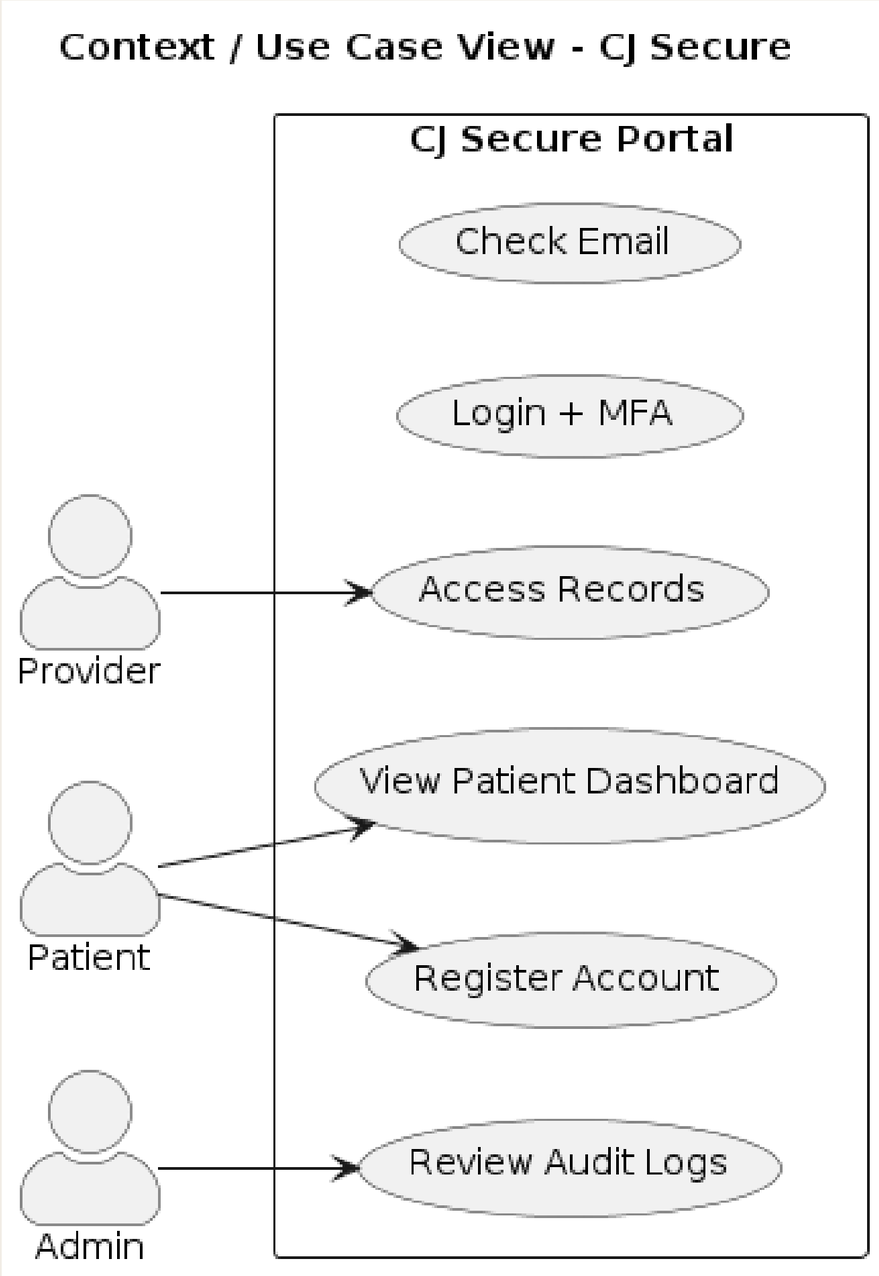


Presented by: Chica Gomes and Janet Griffin

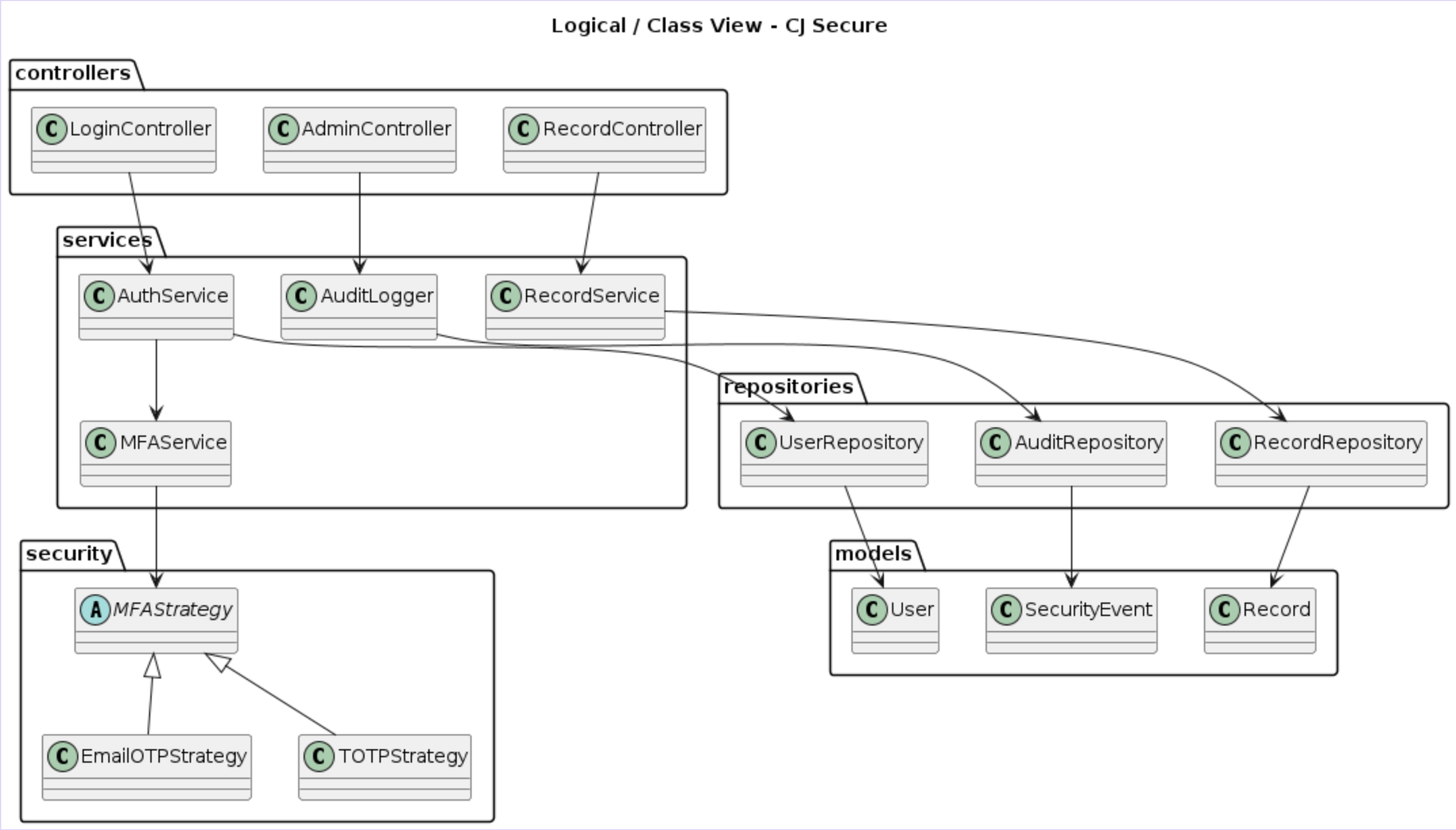
OUR GOAL

- SECURE SENSITIVE HEALTHCARE DATA IN A REALISTIC ENVIRONMENT
- PREVENT UNAUTHORIZED ACCESS USING LAYERED SECURITY (MFA)
- ENSURE ACCOUNTABILITY THROUGH AUDIT LOGGING AND MONITORING

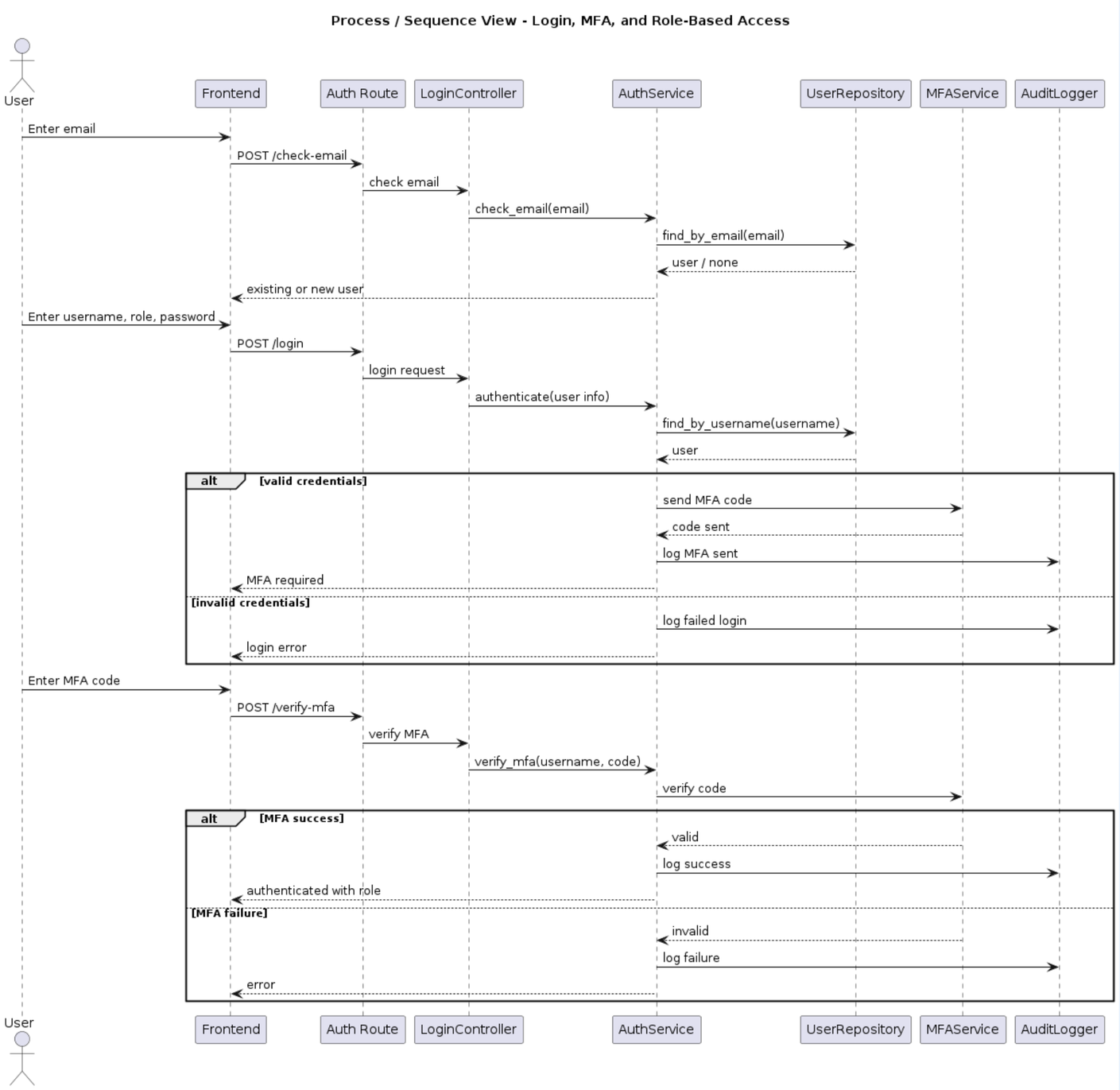
CASE VIEW


Context / Use Case View - CJ Secure
CJ Secure Portal
Check Email
Login + MFA
Access Records
View Patient Dashboard
Register Account
Review Audit Logs
Provider
Patient
Admin

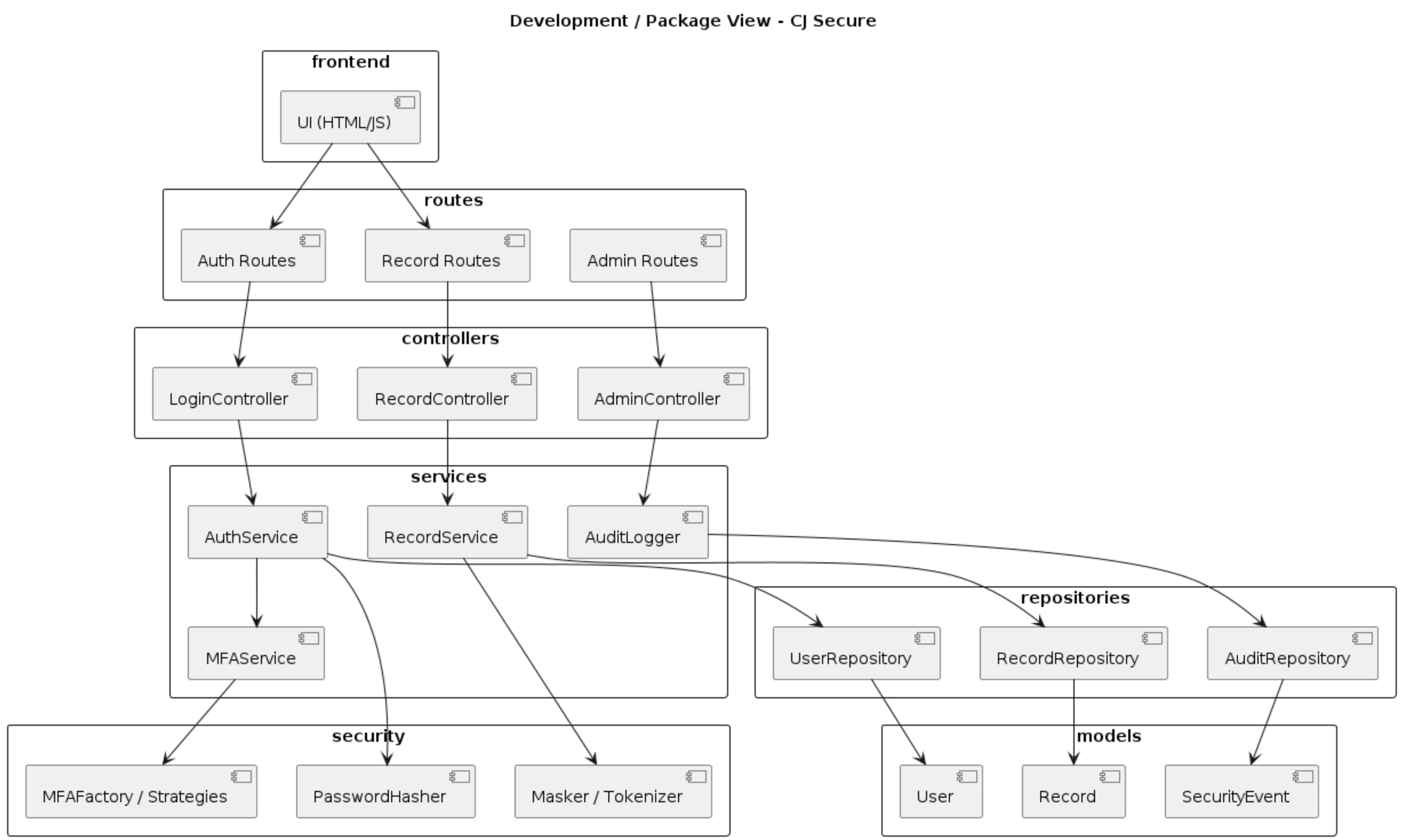
LOGICAL VIEW



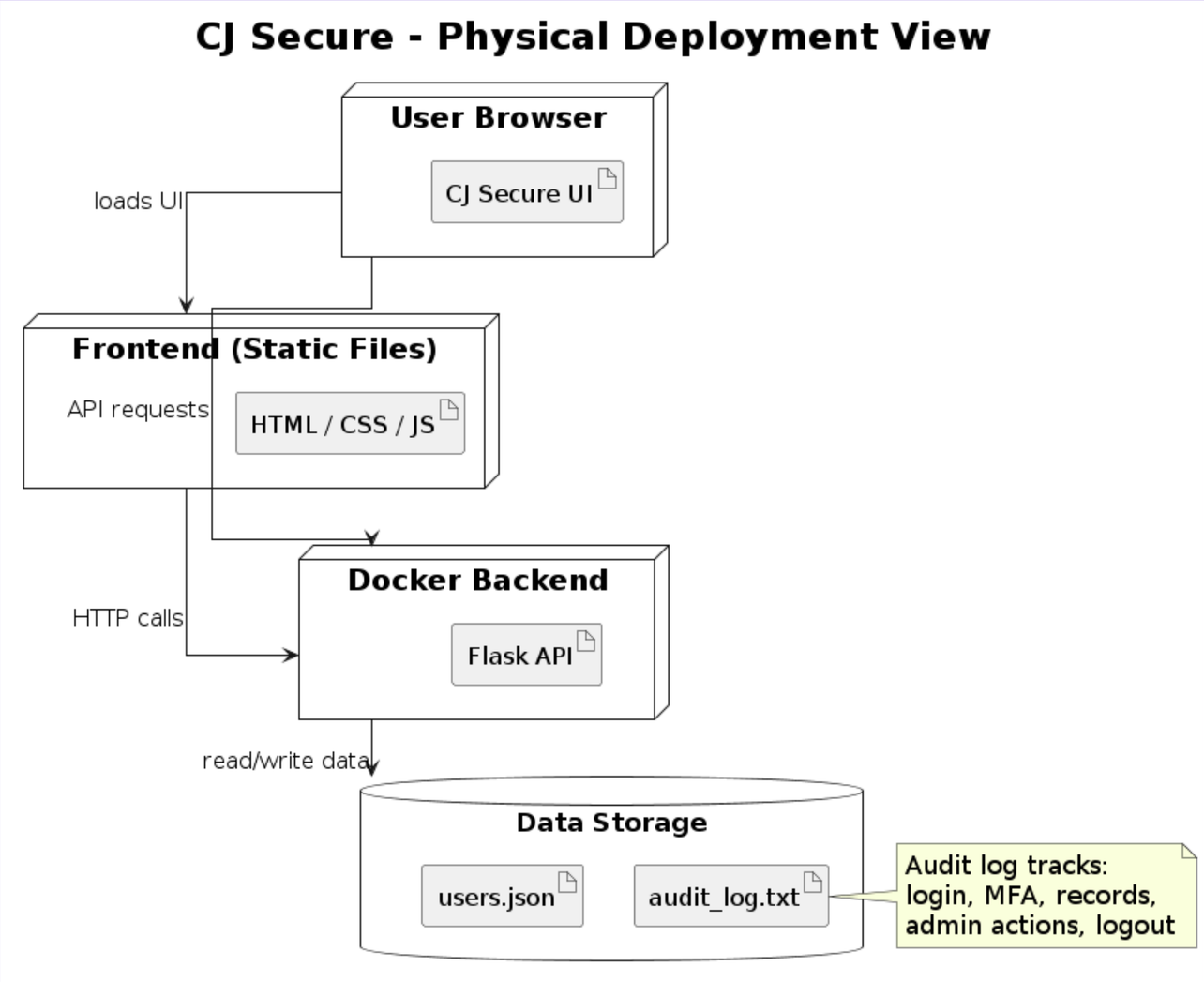
PROCESS SEQUENCE VIEW



DEVELOPMENT VIEW



PHYSICAL DEPLOYMENT VIEW



STRATEGY

```python
class MFAStrategy(ABC):
    """Abstract base class for MFA strategies."""

    @abstractmethod
    def send_code(self, user: Any) -> None:
        """Send an MFA code or prepare an MFA challenge."""
        pass

    @abstractmethod
    def verify_code(self, user: Any, code: str) -> bool:
        """Verify an MFA code."""
        pass
```

FACTORY

```python
class MFAFactory:
    """
    Factory responsible for creating the correct MFA strategy object.
    """

    _email_strategy = EmailOTPStrategy()
    _totp_strategy = TOTPStrategy()

    def create_strategy(self, method: str) -
        if method == "email":
            return self._email_strategy

        if method == "totp":
            return self._totp_strategy

        raise ValueError(f"Unsupported MFA m
```

```python
def __init__(self) -> None:
    """Create the MFA factory."""
    self.mfa_factory = MFAFactory()

def send_mfa_code(self, user, method: str = "email") -> None:
    """
    Send an MFA code using the requested MFA method.
    """
    strategy = self.mfa_factory.create_strategy(method)
    strategy.send_code(user)
```

SINGLETON

```python
def __new__(cls):
    """
    Ensure only one AuditLogger instance is ever created.
    """
    if cls._instance is None:
        cls._instance = super(AuditLogger, cls).__new__(cls)
    return cls._instance

def __init__(self) -> None:
    """
    Initialize the singleton only once.

    Without this guard, every call to AuditLogger() would reset the
    repository and event counter.
    """
    if self.__class__._initialized:
        return

    self.audit_repository = AuditRepository()
    self.next_event_id = 1
    self.__class__._initialized = True
```

DEMONSTRATION